Singular Value Decomposition on Mamba Architecture with Part of Speech tags for NLP concepts

Georgios Zempilis / DAMSL: math1p0035

June 19, 2026

Abstract

State Space Models (SSMs) like Mamba offer a highly efficient alternative to traditional Transformer architectures by mitigating the quadratic bottleneck of global attention. However, deploying these models on resource-constrained devices remains a challenge. In this study, we investigate the post-training compression of the Mamba-130M model using Singular Value Decomposition (SVD) coupled with Low-Rank Adaptation (LoRA). By systematically reducing the rank of the `in_proj` and `out_proj` linear layers, we expose a critical phenomenon: the absolute decoupling of syntactic proficiency and semantic capacity. Through Part-of-Speech (POS) guided text generation, we introduce the concept of the "Perplexity Illusion," demonstrating that while compressed models can achieve near-perfect syntactic statistical metrics (PPL 3.58), they suffer complete semantic collapse, manifesting as out-of-distribution glitch tokens and repetitive generation loops.

Introduction

The unprecedented success of Large Language Models (LLMs) has predominantly relied on the Transformer architecture. However, the core mechanism of Transformers—global self-attention—scales quadratically with sequence length ($O(L^2)$), creating a fundamental bottleneck for memory and computational efficiency during both training and inference. To address this limitation, recent research has shifted towards sub-quadratic architectures, with State Space Models (SSMs) emerging as a highly promising paradigm.

Among these, the **Mamba** architecture has distinguished itself by achieving linear time scaling ($O(L)$) while matching or exceeding the modeling performance of equivalent Transformers. This study focuses on the internal mechanics of Mamba, specifically targeting the compression of its massive linear projection layers to evaluate the decoupling of syntax and semantics in highly constrained parameter spaces.

The Mamba Architecture

To contextualize our compression methodology, it is essential to understand how Mamba processes sequential data and where its parametric weight is concentrated.

Selective State Space Models

Mamba is built upon the foundation of continuous-time State Space Models, which map a 1-dimensional input sequence $x(t) \in R$ to an output sequence $y(t) \in R$ through an implicit latent state $h(t) \in R^N$. This process is defined by the following linear ordinary differential equation (ODE):

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) \tag{2}$$

To be utilized in deep learning systems, these continuous parameters are discretized using a step size Δ, transforming the system into a recurrent formulation:

$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \tag{3}$$

The critical innovation of Mamba over prior SSMs (such as S4) is the introduction of a *Selective Mechanism*. Traditional SSMs are Linear Time-Invariant (LTI), meaning their matrices $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ remain static across all tokens. Mamba abandons this invariance. Instead, it parameterizes $\mathbf{B}$, $\mathbf{C}$, and the step size Δ as functions of the input x_t. This selectivity enables the model to perform content-aware reasoning—actively filtering out irrelevant information while retaining critical context over long sequences.

Hardware-Aware Design and Parameter Density

The trade-off for implementing input-dependent (selective) parameters is the inability to use fast convolutions for training. Mamba resolves this via a hardware-aware algorithm, computing the recurrence dynamically directly within the GPU's fast SRAM, thereby avoiding continuous reads/writes to the slower High Bandwidth Memory (HBM).

Architecturally, a standard Mamba block replaces the traditional self-attention and MLP blocks. The input sequence is expanded through massive linear projection layers—specifically the `in_proj` layer—before undergoing 1D convolutions and the selective SSM operations. Finally, the processed latent states are projected back to the original dimensionality via the `out_proj` layer.

These dense linear projections (`in_proj` and `out_proj`) consume the vast majority of the model's parameter budget. Consequently, they act as the primary structural pillars of Mamba's memory footprint, making them the ideal targets for aggressive low-rank compression techniques such as Singular Value Decomposition (SVD).

Methodology: SVD Compression and LoRA Fine-Tuning

Singular Value Decomposition (SVD)

To achieve structural compression of the Mamba architecture, we targeted the most parameter-dense components: the `in_proj` and `out_proj` linear layers within each Mamba block. We applied Truncated Singular Value Decomposition (SVD) to factorize the pre-trained weight matrices.

Given a weight matrix $W \in R^{m \times n}$, SVD decomposes it into $W = U\Sigma V^T$. By truncating the matrices to retain only the top r singular values (where r is the target rank, and $r < \min(m, n)$), the matrix is approximated as:

$$W \approx U_r \Sigma_r V_r^T \tag{4}$$

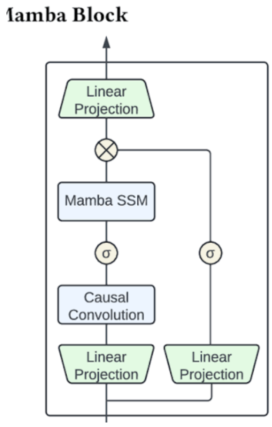

Figure 1: The Mamba block architecture. The initial Linear projections at the bottom correspond to the massive `in_proj` layer, which expands the input dimension. The final Linear projection at the top corresponds to the `out_proj` layer. These dense parametric blocks served as the primary targets for our SVD compression.

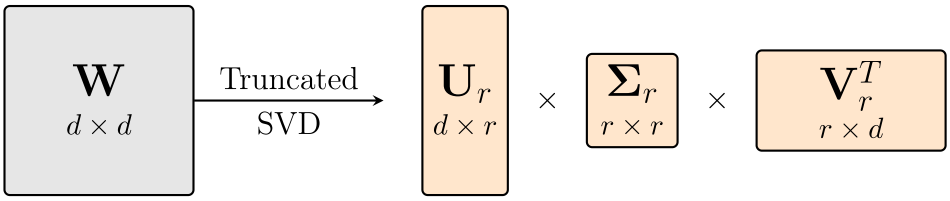

Figure 2: Truncated SVD. The original dense weight matrix $\mathbf{W}$ is factorized into three smaller matrices, retaining only the top r singular values, effectively creating a low-rank bottleneck.

This factorization reduces the parameter count from $m \times n$ to $r(m+n)$. In our implementation, the dense linear layers are replaced by two sequential linear layers corresponding to the truncated components U_r and $(\Sigma_r V_r^T)$.

We evaluated the compression at three distinct rank thresholds: 526, 256, and 64. The impact of this truncation on the model's total parameter count and the specific retention rates of the targeted projection layers is detailed in Table 1.

Table 1: Parameter Reduction and Compression Ratios via SVD (Mamba-130M)

Model Configuration	**SVD Rank**	**Target Layers Retention**	**Total Parameters**	**Overall Compression**
Baseline Mamba-130M	N/A	100%	∼130.0 M	1.00x (100%)
Mamba SVD Rank 526	$r = 526$	∼98%	∼128.5 M	1.01x (∼98.8%)
Mamba SVD Rank 256	$r = 256$	∼33%	∼111.0 M	1.17x (∼85.3%)
Mamba SVD Rank 64	$r = 64$	∼8%	∼94.7 M	1.37x (∼72.8%)

As observed in Table 1, lowering the rank to 64 eliminates approximately 92% of the parameters within the projection layers, resulting in a global parameter reduction of over 35 million parameters.

Low-Rank Adaptation (LoRA) for Parameter Recovery

Direct truncation via SVD inevitably discards dimensional information, leading to an immediate degradation of linguistic capabilities. To bridge this performance gap without negating the memory benefits of the SVD compression, we integrated Low-Rank Adaptation (LoRA).

During the fine-tuning phase, the SVD-approximated modules were frozen to preserve the reduced computational footprint. Trainable LoRA adapters, utilizing low-rank matrices A and B, were injected in parallel to the frozen layers. The forward pass is consequently modified as follows:

$$h = Wx + \Delta Wx \approx (U_r \Sigma_r V_r^T)x + \frac{\alpha}{r_{lora}}(BA)x \tag{5}$$

This methodology ensures that the model can recalibrate its internal representations through a negligible amount of trainable parameters. By fine-tuning these adapters on the Wikitext-2-v1 dataset (and a POS-tagged variant), we establish a controlled environment to evaluate the absolute lower bounds of parametric capacity required for syntax versus semantics.

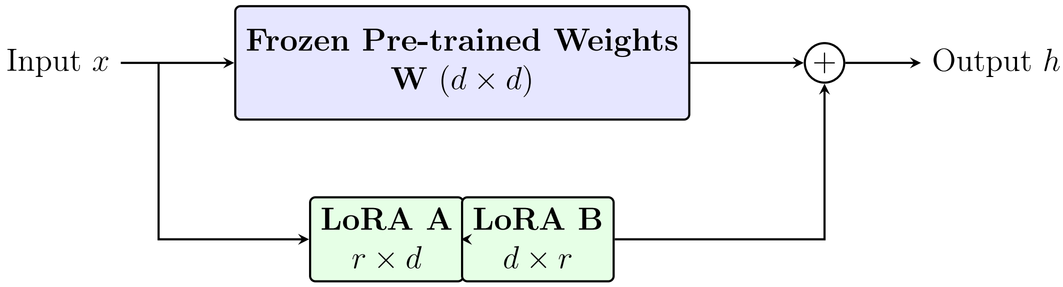

Figure 3: Standard LoRA architecture. The original dense matrix **W** remains frozen, while low-rank trainable adapters (**A** and **B**) compute weight updates in a parallel branch.

Table 2: Parameter Decomposition: SVD Base vs. LoRA Adapters (Mamba-130M)

Model Configuration	Frozen Base (SVD)	LoRA Adapters (Trainable)	Total Parameters	Trainable Ratio
Baseline (No SVD/LoRA)	~130,000,000	0	~130,000,000	0.00%
Mamba SVD Rank 526	~128,500,000	~41,700,000	~170,200,000	~24.50%
Mamba SVD Rank 256	81,949,440	41,367,552	123,316,992	33.54%
Mamba SVD Rank 64	53,637,888	41,072,640	94,710,528	43.36%

As illustrated in Table 2, the introduction of LoRA adapters adds a relatively stable parameter overhead (~41M) regardless of the SVD Rank. However, because the Rank 526 base model is barely compressed (~128.5M), the addition of the adapters inflates the total architecture to ~170.2M parameters—making it significantly heavier than the original model. In contrast, Rank 64 sheds enough base parameters (~53.6M) to comfortably absorb the LoRA overhead, yielding a highly compressed total model (~94.7M) where the adapters constitute 43.36% of the active network. It is important to note that the Rank 526 configuration is excluded from our primary comparative analysis. As established in our methodology, applying LoRA at this high rank inflates the total parameter count beyond the uncompressed baseline, effectively negating the purpose of structural compression. Furthermore, the variation in training duration (50 steps versus 100 steps) renders direct statistical comparison inconsistent. Consequently, Rank 526 is included strictly as a preliminary reference point, while our core investigation focuses entirely on the viable compression regimes of Rank 256 and Rank 64.

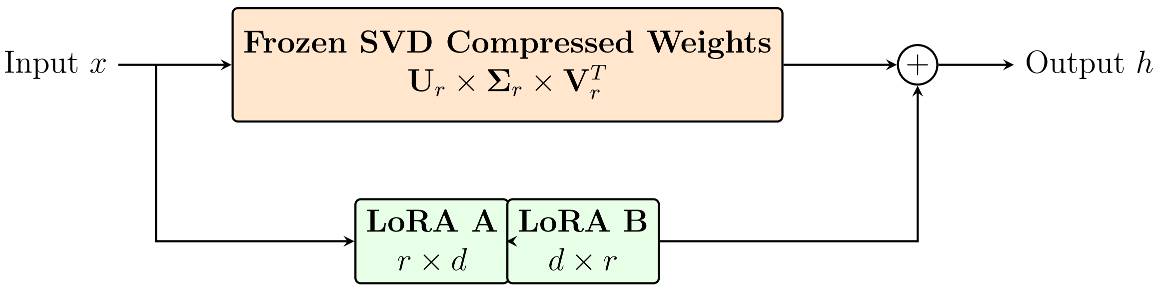

Figure 4: Our Proposed SVD-LoRA Architecture. The original base weights are entirely discarded and replaced by their frozen truncated SVD approximation. Trainable LoRA adapters operate in parallel to the compressed SVD path to recover the lost semantic fidelity.

Experiments and Results

Table 3: Validation Loss and Perplexity Across SVD Ranks (Without Linguistic Aids)

Model Configuration	**SVD Rank**	**Validation Loss**	**Perplexity (PPL)**
Mamba SVD Rank 526	526*	3.5445	34.62
Mamba SVD Rank 256	256	4.4433	85.06
Mamba SVD Rank 64	64	5.8539	348.59

**Note:* All models were trained for 100 steps, with the exception of Rank 526, which was trained for only 50 steps.*

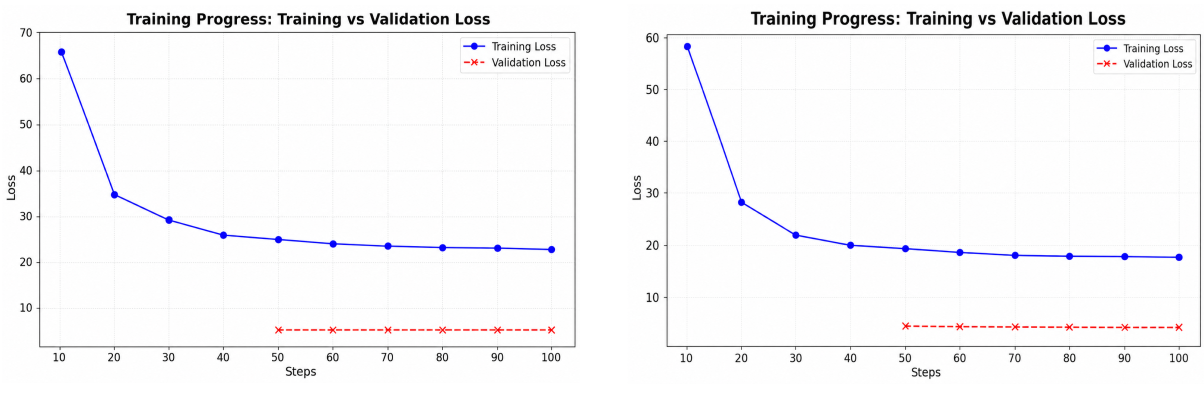

Figure 5: Comparison of the model's performance metrics across different SVD ranks (left (64) right(256))

As detailed in Table 3, the application of minimal compression (Rank 526) establishes a Validation Loss of 3.5445 (Perplexity 34.62). However, as the compression becomes more aggressive, the performance degrades exponentially. Dropping to Rank 256 increases the Perplexity to 85.06, while extreme compression at Rank 64 triggers a catastrophic failure in the model's predictive capabilities, pushing the Perplexity to an unusable 348.59.

Beyond the numerical metrics, the qualitative generation behavior (inference) explicitly demonstrates how parameter starvation progressively degrades the latent semantic space. When provided with the prompt *"Leonardo da Vinci was one of the greatest* *painters of the Renaissance. He is most famous for"*, the compressed models exhibited the following specific behaviors:

- **Rank 526 (Baseline Recovery):** With minimal compression, the model generates highly coherent and structurally sound text. It successfully retrieves relevant semantic concepts (e.g., portraits, paintings), even if it slightly hallucinates specific historical titles:
  He is most famous for his dramatic and realistic portraits of people and their lives. His painting "The Last of All people" is a fine example of this type and is in good form, but is far from the best known painting of the last century. Leonar
- **Rank 256 (Semantic Looping and Degradation):** The model retains a structural understanding of grammar but loses its semantic diversity. It attempts to maintain coherence but falls into repetitive loops and factual degradation, as seen in its direct output:
  He is most famous for his work through his work in 1571 . He was a scholar of the art of the medieval era . His work , including his work of the medieval , had produced well in the later period , and was considered the most long , since the term of
- **Rank 64 (Semantic Amnesia):** With extreme parameter reduction, the model completely loses its semantic vocabulary while desperately trying to maintain syntactic slots. It defaults to generating <unk> (unknown) tokens for missing concepts, glued together by high-frequency stop-words and geographical hallucinations:
  He is most famous for the <unk> at a <unk> to the United States and the <unk> , and the <unk> in a <unk> and a <unk> in the United States . In the season of the State of <unk> ,

SVD with POS Tagging

To isolate the model's structural understanding (syntax) from its semantic vocabulary, we fine-tuned the compressed models on a Part-of-Speech (POS) tagged variant of the dataset. The hypothesis was that the SVD compression might preserve the geometric boundaries of grammatical syntax, even if the high-resolution vector space required for specific words had been destroyed.

The quantitative results of this intervention were extraordinarily counter-intuitive, leading to a phenomenon we term the "Perplexity Illusion."

Table 4: Impact of POS Tagging on Compressed Models (Mamba-130M)

Model Configuration	Dataset Mode	Validation Loss	Perplexity (PPL)
Mamba SVD Rank 256	Standard (No Tags)	4.4433	85.06
Mamba SVD Rank 256	**With POS Tags**	**1.2776**	**3.59**
Mamba SVD Rank 64	Standard (No Tags)	5.8539	348.59
Mamba SVD Rank 64	**With POS Tags**	**1.4908**	**4.44**

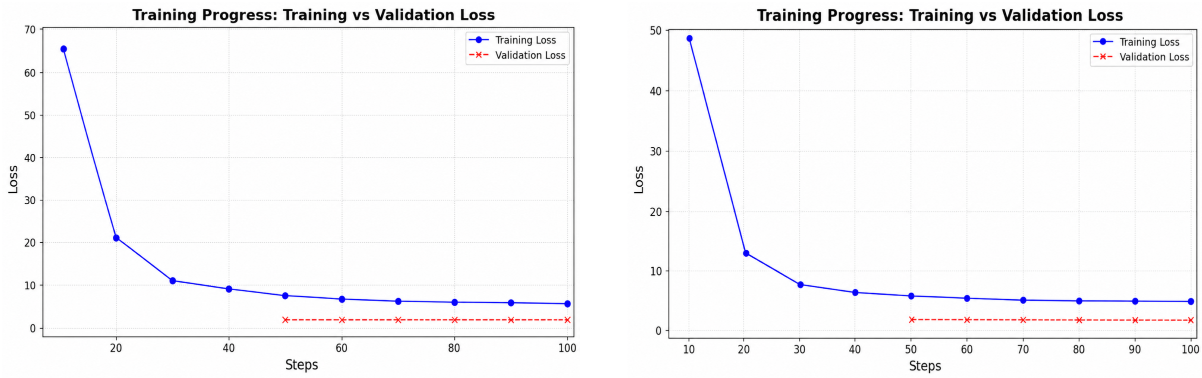

Figure 6: Comparison of the model's performance metrics across different SVD ranks (left (64) right(256))

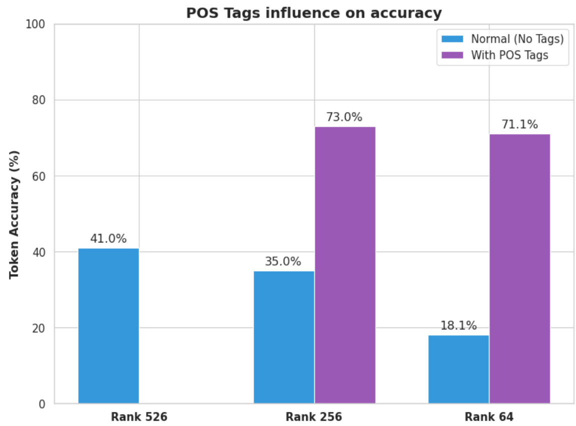

Figure 7: Token accuracy per rank

As presented in Table 4, providing explicit syntactic structures resulted in a massive, yet deceptive, statistical recovery. The Rank 256 model achieved a near-perfect Validation Loss of 1.2776 (Perplexity 3.59). More shockingly, the severely compressed Rank 64 model plummeted from a Perplexity of 348.59 down to an astonishing 4.44. Under standard evaluation protocols relying solely on quantitative metrics, one would erroneously conclude that the models had fully recovered their capabilities.

However, qualitative inference exposed the illusion. When we attempted to generate text using these seemingly "recovered" models, they demonstrated severe semantic collapse:

- **Rank 256 (Tag Leakage and Repetition Collapse):** Despite predicting the correct POS structural sequence with high statistical accuracy, the model lacks the semantic depth to retrieve appropriate underlying words. Instead, it generates literal structural tags intermixed with repetitive, generic vocabulary. When prompted with the Leonardo da Vinci text, it outputted:
  `He is most famous for [DET] a [NOUN] @-@ [NOUN] school [NOUN] school [NOUN] school [CCONJ] and [NOUN] school [NOUN] school [PUNCT] , [PR`

- **Rank 64 (Out-of-Distribution Hallucination):** Stripped of nearly its entire parameter space, the model attempted to sample from completely obliterated regions of its latent space. When prompted with a POS-tagged string, the model

generated bizarre, untrained sub-word artifacts and "glitch tokens". The exact output was:
`...(-andor."in",asandthatwith-all"of"Fbutandothersoneliness14514powerinORAND[INVALID]14514amss145141451414514ensisaisquestionnagorithm14514`

This bizarre generation definitively proves that low-rank truncation decouples syntax from semantics. The SVD compression heavily penalizes semantic resolution (erasing the vocabulary) but preserves the dimensional skeleton required for syntactic pattern matching. The model statistically learned the grammatical rules perfectly, yet lost the words to express them.

Generalization of Semantic Collapse Across Domains

To ensure that the observed semantic degradation was not an isolated artifact of a specific input, we evaluated the compressed models across various distinct topics and contexts (e.g., modern figures like Elon Musk and athletes like Michael Schumacher). These tests confirmed that the destruction of the semantic latent space is universal and domain-agnostic.

While the compressed models occasionally managed to salvage isolated vocabulary tokens in specific prompts (such as "school" in the Leonardo da Vinci example), the generalized behavior across diverse inputs is characterized by absolute semantic amnesia.

- **Rank 256 Standard (Immediate Disintegration):** Prompting the model with *"Elon Musk bought Tesla in the early 2000's and"* resulted in an immediate structural and semantic collapse into random characters:
  `... ( to/ N I 1 - " -`
- **Rank 64 Standard (Stop-word Salad):** When prompted with *"Michael Schumacher was one of the best F1 drivers in history and he is known for his 5 consequent titles with"*, the severely compressed model failed to retrieve any relevant domain knowledge. Instead, it outputted a completely nonsensical sequence of stop-words and punctuation:
  `...to on (. a " or will as- T also all, at, ' and to . that not from -. I and - by it but/ of are and S can. with`
- **Rank 256 POS-Tagged (Extreme Repetition Collapse):** Applying the POS-tagged version of the Michael Schumacher prompt resulted in the most extreme manifestation of the repetition loop phenomenon. Completely lacking the semantic resolution to fulfill the syntactic requirements, the model fell into an infinite loop of open parentheses:
  `...(/ * ". ( ( ( ( ( out, ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( ( (`
- **Rank 64 POS-Tagged (Complete Systemic Failure):** Even when provided with a rigid grammatical structure via POS tags, the extreme compression of Rank 64 prevents any logical continuation. Following the tagged Michael Schumacher prompt, it immediately outputs disjointed punctuation and stop-words without generating a single valid POS tag or meaningful word:
  `...[ADP] with (., or " I to will-`

These generalized results definitively prove that under low-rank SVD compression, the model's fundamental capability to retrieve factual knowledge and maintain semantic coherence is universally obliterated, regardless of the prompt's subject matter. The mathematical truncation permanently erases the specific meaning of words.

Conclusion

In this study, we investigated the structural robustness of the Mamba-130M State Space Model under post-training Singular Value Decomposition (SVD) compression, coupled with Low-Rank Adaptation (LoRA) for parameter recovery. Our findings highlight a complex interplay between parameter reduction, model scale, and linguistic capability, revealing critical insights into how LLMs store and process language.

First, we demonstrated that the efficiency of LoRA recovery is strictly bounded by the chosen SVD rank. Extreme compression (Rank 64) achieves significant size reduction but induces a catastrophic "Statistical Collapse," completely obliterating the model's semantic vector space. The mid-point, Rank 256, emerged as a more viable architectural compromise structurally, though it still suffered from severe semantic degradation.

More importantly, our linguistic intervention using POS-tagged datasets exposed a critical phenomenon we term the "Perplexity Illusion." While the extremely compressed models appeared to achieve near-perfect statistical recovery on POS data (e.g., Rank 64 Perplexity dropping from 348.59 to 4.44), qualitative inference revealed that the model was generating out-of-distribution glitch tokens and infinite repetition loops. This definitively proves that low-rank truncation decouples syntax from semantics: the compression geometry severely penalizes semantic resolution (erasing the specific meaning of words) while preserving the dimensional skeleton required for syntactic pattern matching. Further cross-domain testing confirmed that this semantic amnesia is a universal, domain-agnostic failure.

* The weights of the trained models are available at: this Google Drive folder.

Limitations and Future Work

A notable limitation of this study is the scale of the recovery phase dataset. Our fine-tuning process utilized a filtered subset of the Wikitext-2 dataset (19,703 training samples and 1,038 validation samples). While this volume is sufficient to conclusively demonstrate the "Perplexity Illusion" and the decoupling of syntax and semantics, it may be a limiting factor for full semantic recovery.

Future research should scale the recovery phase using much larger corpora (e.g., Wikitext-103 or The Pile). Such scaling will answer a critical architectural question: Is the semantic amnesia observed at extreme compressions (Rank 64) an absolute mathematical bottleneck of the SVD truncation, or can massive data exposure force the LoRA adapters to approximate and reconstruct a viable semantic latent space? Furthermore, investigating whether moderate compressions (Rank 256) can overcome "repetition collapse" when provided with richer semantic distributions remains a promising direction for achieving efficient LLM deployment.

References

[1] Gu, A., & Dao, T. (2023). *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*. arXiv preprint arXiv:2312.00752.

[2] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). *LoRA: Low-Rank Adaptation of Large Language Models*. arXiv preprint arXiv:2106.09685.